

Spectral Shape Analysis for Electron Antineutrino Oscillation Study by Using ^8Li generator with ^{252}Cf Source

Source

This article focuses on a method of detecting antineutrinos from a sample of ^8Li and how this method could lead to evidence for sterile neutrinos. This proposed method is short-baseline and utilizes a generator of ^8Li so that no accelerator or reactor is needed. ^{252}Cf undergoes neutron capture to ^7Li , creating ^8Li in the process. The actual antineutrino detection, however, occurs by inverse beta decay, as prompt signal and a delayed signal are observed- almost definite signs of an IBD. A layer of graphite is placed both between the ^{252}Cf and ^7Li layers and acts as a modulator or a reflector source. Since graphite has a low absorption cross-section and a high elastic scattering cross-section for neutrons, this setup helps produce more ^8Li . The cross-sectional diagrams of both the generator and the detector from the paper are shown below:

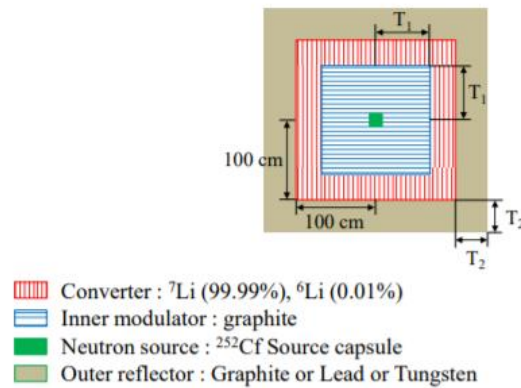


Figure 1. (Color online) A schematic cross section view showing cylindrical ^8Li generator.

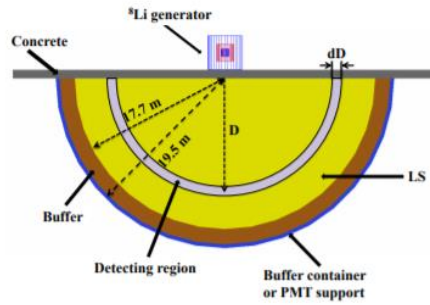


Figure 2. (Color online) Simulation geometry for hemisphere shape detector with ^8Li generator.

The authors then explain their logic behind the designs of the generator and reactor and delve into how they chose the best proportions for this experiment. Given the materials used, the authors find a yield of $0.0045 \text{ } ^8\text{Li}$ per neutron. They then discuss figuring out an appropriate most efficient thickness for the graphite in both parts of the generator. While thicker graphite leads to more neutron/carbon nucleus collisions, this layer of graphite will make it difficult for neutrons to escape when too thick. A maximum $^8\text{Li}/n$ yield of 0.256 occurs when $T_1=43\text{cm}$ and $T_2=100\text{cm}$, so both are used in the generator. The authors then discuss the presence of background antineutrinos and conclude that the other isotopes in this setup have low yields and relatively long half-lives and therefore contribute very few events. The effects of neutrinos from ^{252}Cf end up being three orders of magnitude smaller than those from the Li present so they can also be considered as negligible. And finally, the energy cuts applied to this system make it so that the energy of geoneutrinos falls below them, so they also are not taken into account.

Finally, the authors analyze the spectral shape of the neutrino oscillations from the detector and initially find that the shape of the energy oscillations changes as the distance of the detectors, D in figure 2, changes. They look at the ratios of the expected total event ratio of both the $3+1$ and $3+2$ sterile neutrino models to the total expected total event ratio. It is observed that the peak position of neutrino energies increases as D increases for the $3+1$ model, but not the $3+2$ model. When $D=8\text{m}$ for

the 3+1 model, the ratios at 12.5MeV and 7.5MeV are 0.917 and 0.988, respectively. However, when $D=14.5\text{m}$, these ratios change to 0.994 and 0.922. The authors then conclude that a 15% deviation from the expected measurements will show if the 3+1 sterile neutrino model is the best model to observe for sterile neutrino discovery.

Overall, I would publish this paper with major revisions. I appreciate this experiment and its potential contributions to the search for sterile neutrinos, but I am honestly disappointed that there are not more experimental results. The few results the authors mention are tacked onto the end of the paper and definitely do not feel like its main focus.